ARTIFICIAL INTELLIGENCE

U25AD301 | Question Bank – Units I to V
PART A – 1 MARK MULTIPLE-CHOICE QUESTIONS WITH ANSWERS

Based on the uploaded KPR Institute of Engineering and Technology Question Bank (2026–2027).

UNIT I – INTRODUCTION TO ARTIFICIAL INTELLIGENCE

Total: 20 questions

1. Which of the following is a branch of Artificial Intelligence?

a) Machine Learning
b) Compiler Design
c) Operating System
d) DBMS
Correct Answer: a) Machine Learning

2. Which component enables an intelligent agent to perceive its environment?

a) Actuator
b) Performance Measure
c) Agent Program
d) Sensor
Correct Answer: d) Sensor

3. Which component maps percept sequences to actions?

a) Agent Function
b) Performance Measure
c) Environment
d) Sensor
Correct Answer: a) Agent Function

4. A rational agent selects an action that

a) Produces random results
b) Minimizes computation only
c) Maximizes the expected performance measure
d) Always reaches the goal in one step
Correct Answer: c) Maximizes the expected performance measure

5. PEAS stands for

a) Performance Measure, Environment, Actuators, Sensors
b) Performance, Environment, Agents, Systems
c) Process, Environment, Actuators, Sensors
d) Process, Event, Actuators, Sensors
Correct Answer: a) Performance Measure, Environment, Actuators, Sensors

6. A robotic vacuum cleaner moves between rooms and cleans dirty locations. Which type of environment best describes this problem?

a) Continuous
b) Partially Observable
c) Single-Agent
d) Discrete
Correct Answer: d) Discrete

7. Which of the following is an example of a fully observable environment?

a) Chess
b) Poker
c) Taxi Driving
d) Medical Diagnosis
Correct Answer: a) Chess

8. Which type of intelligent agent uses condition–action rules?

a) Simple Reflex Agent
b) Goal-Based Agent
c) Utility-Based Agent
d) Learning Agent
Correct Answer: a) Simple Reflex Agent

9. A mobile robot stores information about obstacles that are not currently visible. Which agent architecture does it use?

a) Simple Reflex Agent
b) Model-Based Reflex Agent
c) Utility-Based Agent
d) Learning Agent
Correct Answer: b) Model-Based Reflex Agent

10. A utility-based agent selects an action based on

a) Random choice
b) Fixed rules only
c) Maximum utility value
d) Number of sensors
Correct Answer: c) Maximum utility value

11. Which agent improves its performance through experience?

a) Learning Agent
b) Random Agent
c) Reflex Agent
d) Search Agent
Correct Answer: a) Learning Agent

12. Which component is responsible for perceiving the environment?

a) Actuator
b) Sensor
c) Agent Program
d) Performance Measure
Correct Answer: b) Sensor

13. Which of the following is an actuator?

a) Camera
b) Keyboard
c) Motor
d) Microphone
Correct Answer: c) Motor

14. A robot maintains an internal model of its environment to update changes before selecting an action. Which type of intelligent agent best represents this behavior?

a) Simple Reflex Agent
b) Goal-Based Agent
c) Model-Based Reflex Agent
d) Learning Agent
Correct Answer: c) Model-Based Reflex Agent

15. An autonomous drone uses GPS, cameras, and motors to reach its destination. Which of the following is an actuator?

a) Camera
b) GPS
c) Motor
d) Sensor
Correct Answer: c) Motor

16. In a deterministic environment, an action always produces

a) Random results
b) Multiple possible outcomes
c) No outcome
d) A predictable outcome
Correct Answer: d) A predictable outcome

17. Which agent architecture employs a utility function for decision-making? (GATE-2024)

a) Goal-Based Agent
b) Utility-Based Agent
c) Simple Reflex Agent
d) Learning Agent
Correct Answer: b) Utility-Based Agent

18. Which search strategy expands the shallowest node first? (GATE-2023)

a) Depth First Search
b) Breadth First Search
c) Uniform Cost Search
d) Depth Limited Search
Correct Answer: b) Breadth First Search

19. Which AI application extensively uses intelligent agents?

a) Autonomous Vehicles
b) Calculator
c) Text Editor
d) Spreadsheet
Correct Answer: a) Autonomous Vehicles

20. The effectiveness of a rational agent is measured using the

a) Utility Function
b) Performance Measure
c) Agent Function
d) Environment
Correct Answer: b) Performance Measure

UNIT II – INFORMED AND LOCAL SEARCH

Total: 20 questions

1. Which search algorithm uses both the path cost and heuristic function to determine the optimal solution?

a) Breadth First Search
b) Depth First Search
c) A* Search
d) Greedy Best-First Search
Correct Answer: c) A* Search

2. Which search algorithm selects the next node based only on heuristic values?

a) Uniform Cost Search
b) Greedy Best-First Search
c) Breadth First Search
d) Depth Limited Search
Correct Answer: b) Greedy Best-First Search

3. Which evaluation function is used in the A* search algorithm?

a) h(n)
b) g(n) – h(n)
c) g(n)
d) g(n) + h(n)
Correct Answer: d) g(n) + h(n)

4. Which type of heuristic guarantees that A* Search finds an optimal solution?

a) Inadmissible heuristic
b) Admissible heuristic
c) Random heuristic
d) Dynamic heuristic
Correct Answer: b) Admissible heuristic

5. Which search algorithm is commonly used to solve route-finding problems?

a) Breadth First Search
b) A* Search
c) Depth First Search
d) Iterative Deepening Search
Correct Answer: b) A* Search

6. Which local search algorithm may become trapped at a local optimum?

a) Simulated Annealing
b) Genetic Algorithm
c) Hill Climbing
d) A* Search
Correct Answer: c) Hill Climbing

7. Which optimization algorithm accepts worse solutions with a certain probability to escape local optima?

a) Hill Climbing
b) Greedy Best-First Search
c) Simulated Annealing
d) Breadth First Search
Correct Answer: c) Simulated Annealing

8. Which Genetic Algorithm operator combines two parent chromosomes to produce offspring?

a) Mutation
b) Selection
c) Crossover
d) Evaluation
Correct Answer: c) Crossover

9. Which search technique is suitable for solving optimization problems with large search spaces?

a) Local Search
b) Breadth First Search
c) Uniform Cost Search
d) Depth First Search
Correct Answer: a) Local Search

10. Hill Climbing always produces the global optimum solution.

a) True
b) False
c) Only for heuristic search
d) Only for deterministic problems
Correct Answer: b) False

11. Which Genetic Algorithm operator introduces random changes into a chromosome?

a) Mutation
b) Selection
c) Crossover
d) Fitness Evaluation
Correct Answer: a) Mutation

12. Which parameter controls the probability of accepting worse solutions in Simulated Annealing?

a) Population Size
b) Temperature
c) Fitness Value
d) Mutation Rate
Correct Answer: b) Temperature

13. Which search algorithm generally expands fewer nodes by using heuristic information?

a) Breadth First Search
b) Depth First Search
c) A* Search
d) Uniform Cost Search
Correct Answer: c) A* Search

14. A search algorithm expands the node having the smallest heuristic value without considering the path cost. Which algorithm is used?

a) Hill Climbing
b) Greedy Best-First Search
c) A* Search
d) Uniform Cost Search
Correct Answer: b) Greedy Best-First Search

15. Which component of a Genetic Algorithm evaluates the quality of a solution?

a) Mutation
b) Selection
c) Crossover
d) Fitness Evaluation
Correct Answer: d) Fitness Evaluation

16. Simulated Annealing is more effective than Hill Climbing in escaping local optima.

a) True
b) False
c) Only for uninformed search
d) Only for graph search
Correct Answer: a) True

17. Which search algorithm uses the evaluation function f(n)=g(n)+h(n)? (GATE-2024)

a) Hill Climbing
b) Greedy Best-First Search
c) A* Search
d) Uniform Cost Search
Correct Answer: c) A* Search

18. Which optimization algorithm uses a cooling schedule during the search process?

a) Hill Climbing
b) Simulated Annealing
c) Genetic Algorithm
d) Breadth First Search
Correct Answer: b) Simulated Annealing

19. Which optimization algorithm is widely used to solve the Traveling Salesman Problem?

a) Breadth First Search
b) Depth First Search
c) Genetic Algorithm
d) Uniform Cost Search
Correct Answer: c) Genetic Algorithm

20. Which optimization technique maintains a population of candidate solutions during the search process? (GATE-2024)

a) Hill Climbing
b) Greedy Best-First Search
c) Simulated Annealing
d) Genetic Algorithm
Correct Answer: d) Genetic Algorithm

UNIT III – CSP AND ADVERSARIAL SEARCH

Total: 20 questions

1. Why is Minimax suitable for two-player games?

a) It uses random moves
b) It assumes both players play optimally
c) It ignores the opponent
d) It searches only one level
Correct Answer: b) It assumes both players play optimally

2. Which condition allows Alpha-Beta pruning to eliminate a subtree?

a) Goal node is reached
b) The subtree cannot improve the final decision
c) Maximum depth is reached
d) All nodes are expanded
Correct Answer: b) The subtree cannot improve the final decision

3. Why does Alpha-Beta pruning produce the same result as Minimax?

a) It evaluates fewer leaf nodes without changing the optimal decision
b) It changes the utility values
c) It ignores MAX nodes
d) It always explores all nodes
Correct Answer: a) It evaluates fewer leaf nodes without changing the optimal decision

4. Why are stochastic games different from deterministic games?

a) They have only one player
b) Their outcomes involve uncertainty
c) They do not require search
d) They have no utility values
Correct Answer: b) Their outcomes involve uncertainty

5. Which characteristic makes Poker a partially observable game?

a) Fixed rules
b) Hidden cards of opponents
c) Two players
d) Turn-based moves
Correct Answer: b) Hidden cards of opponents

6. Why is constraint propagation applied before backtracking?

a) To increase the search space
b) To reduce inconsistent variable assignments
c) To remove variables
d) To generate random solutions
Correct Answer: b) To reduce inconsistent variable assignments

7. Which factor mainly determines the efficiency of a CSP solution?

a) Number of variables only
b) Number of constraints only
c) Variables, domains, and constraints together
d) Number of goals
Correct Answer: c) Variables, domains, and constraints together

8. Why is backtracking considered a complete search algorithm?

a) It explores all possible assignments if necessary
b) It always finds the shortest path
c) It uses randomness
d) It ignores constraints
Correct Answer: a) It explores all possible assignments if necessary

9. Which situation is most suitable for Local Search?

a) Small search space
b) Large optimization problems
c) Tree traversal
d) Binary search
Correct Answer: b) Large optimization problems

10. Why is Minimax not directly applicable to stochastic games?

a) It assumes deterministic outcomes
b) It uses probability
c) It ignores utility values
d) It works only for CSPs
Correct Answer: a) It assumes deterministic outcomes

11. Which statement best differentiates CSP from ordinary search?

a) CSP represents problems using variables, domains and constraints
b) CSP ignores variables
c) CSP has no goal state
d) CSP does not use search
Correct Answer: a) CSP represents problems using variables, domains and constraints

12. Which factor reduces the number of nodes expanded in Alpha-Beta pruning?

a) Good move ordering
b) Increasing depth
c) More utility values
d) Random exploration
Correct Answer: a) Good move ordering

13. Why is a game tree evaluated from leaf nodes to the root?

a) Utility values are known only at terminal states
b) Root node has maximum value
c) Search starts from leaves
d) It reduces memory
Correct Answer: a) Utility values are known only at terminal states

14. Which problem is naturally modeled as a Constraint Satisfaction Problem?

a) Sudoku
b) Binary Search
c) Merge Sort
d) Stack Implementation
Correct Answer: a) Sudoku

15. What is the primary advantage of constraint propagation?

a) It removes impossible values before search
b) It increases variable domains
c) It generates random solutions
d) It removes constraints
Correct Answer: a) It removes impossible values before search

16. Why is Local Search sometimes unable to find the global optimum?

a) It may get trapped in local optima
b) It ignores variables
c) It always terminates immediately
d) It explores every solution
Correct Answer: a) It may get trapped in local optima

17. In a game tree, why does the MIN player choose the minimum utility value?

a) To minimize the MAX player's advantage
b) To increase search depth
c) To maximize its own branching factor
d) To reduce memory usage
Correct Answer: a) To minimize the MAX player's advantage

18. Which feature distinguishes partially observable games from fully observable games?

a) Some information about the environment is hidden
b) They have no utility values
c) They contain only one player
d) They are always deterministic
Correct Answer: a) Some information about the environment is hidden

19. Why is CSP represented using variables, domains, and constraints?

a) To systematically model relationships among possible assignments
b) To reduce memory only
c) To eliminate search
d) To increase branching factor
Correct Answer: a) To systematically model relationships among possible assignments

20. Which statement best explains why Alpha-Beta pruning is preferred over Minimax?

a) It finds the same optimal solution while evaluating fewer nodes
b) It changes the game strategy
c) It ignores MIN nodes
d) It always searches to a smaller depth
Correct Answer: a) It finds the same optimal solution while evaluating fewer nodes

UNIT IV – KNOWLEDGE REPRESENTATION AND LOGIC

Total: 20 questions

1. Which component stores facts and rules in a knowledge-based system?

a) Knowledge Base
b) Knowledge Base
c) Agent Program
d) Environment
Correct Answer: a) Knowledge Base

2. Which type of agent uses stored knowledge to make decisions?

a) Reflex Agent
b) Learning Agent
c) Knowledge-Based Agent
d) Utility-Based Agent
Correct Answer: c) Knowledge-Based Agent

3. Which logic represents knowledge using propositions that are either true or false?

a) Predicate Logic
b) First-Order Logic
c) Propositional Logic
d) Fuzzy Logic
Correct Answer: c) Propositional Logic

4. Which logical connective represents implication?

a) $\wedge$
b) $\vee$
c) $\rightarrow$
d) $\neg$
Correct Answer: c) $\rightarrow$

5. Which normal form is commonly used in the Resolution algorithm?

a) DNF
b) CNF
c) BNF
d) PNF
Correct Answer: b) CNF

6. Which inference rule derives the conclusion Q from $P \rightarrow Q$ and P?

a) Modus Tollens
b) Resolution
c) Modus Ponens
d) Universal Instantiation
Correct Answer: c) Modus Ponens

7. Why is a propositional logic expression converted into CNF before applying Resolution?

a) To simplify logical inference
b) To increase memory usage
c) To remove predicates
d) To simplify arithmetic operations
Correct Answer: a) To simplify logical inference

8. Which inference technique derives conclusions by combining clauses containing complementary literals?

a) Forward Chaining
b) Backward Chaining
c) Resolution
d) Unification
Correct Answer: c) Resolution

9. Which logic supports variables, predicates, and quantifiers?

a) Propositional Logic
b) First-Order Logic
c) Boolean Logic
d) Modal Logic
Correct Answer: b) First-Order Logic

10. First-Order Logic is more expressive than Propositional Logic.

a) True
b) False
c) Only for Horn clauses
d) Only for Resolution
Correct Answer: a) True

11. Which quantifier represents 'for all'?

a) $\exists$
b) $\forall$
c) $\wedge$
d) $\rightarrow$
Correct Answer: b) $\forall$

12. Which quantifier represents 'there exists'?

a) $\forall$
b) $\exists$
c) $\neg$
d) $\vee$
Correct Answer: b) $\exists$

13. Which inference method starts with known facts and derives new facts until the goal is reached?

a) Backward Chaining
b) Resolution
c) Forward Chaining
d) Unification
Correct Answer: c) Forward Chaining

14. Which inference technique starts from the goal and works backward to prove it?

a) Backward Chaining
b) Resolution
c) Forward Chaining
d) Modus Ponens
Correct Answer: a) Backward Chaining

15. Which component performs logical reasoning in a knowledge-based system?

a) Knowledge Base
b) Inference Engine
c) Actuator
d) Sensor
Correct Answer: b) Inference Engine

16. Apply Modus Ponens to the statements $P \rightarrow Q$ and P. What is the conclusion? (GATE-2024)

a) P
b) Q
c) $\neg Q$
d) $\neg P$
Correct Answer: b) Q

17. Which normal form is required before applying the Resolution algorithm? (GATE-2023)

a) DNF
b) CNF
c) BNF
d) PNF
Correct Answer: b) CNF

18. Which inference method begins reasoning from the goal state?

a) Forward Chaining
b) Resolution
c) Backward Chaining
d) Modus Ponens
Correct Answer: c) Backward Chaining

19. Which logic is commonly used to represent relationships among objects in Artificial Intelligence?

a) Propositional Logic
b) Boolean Logic
c) First-Order Logic
d) Temporal Logic
Correct Answer: c) First-Order Logic

20. Which inference technique is commonly used to prove logical conclusions in automated reasoning?

a) Resolution
b) Selection
c) Mutation
d) Hill Climbing
Correct Answer: a) Resolution

UNIT V – UNCERTAINTY, BAYESIAN NETWORKS AND ONTOLOGIES

Total: 20 questions

1. Which concept enables an AI system to make decisions under uncertain conditions?

a) Deterministic reasoning
b) Uncertainty reasoning
c) Boolean reasoning
d) Linear reasoning
Correct Answer: b) Uncertainty reasoning

2. Which probabilistic model represents conditional dependencies among variables?

a) Decision Tree
b) Bayesian Network
c) Neural Network
d) State Space Graph
Correct Answer: b) Bayesian Network

3. Which theorem forms the basis of Bayesian Networks?

a) Bayes' Theorem
b) De Morgan's Law
c) Resolution Principle
d) Modus Ponens
Correct Answer: a) Bayes' Theorem

4. What is the primary purpose of Bayesian Networks in Artificial Intelligence?

a) Knowledge representation under uncertainty
b) Image compression
c) Database normalization
d) Compiler optimization
Correct Answer: a) Knowledge representation under uncertainty

5. Monte Carlo Search primarily uses

a) Deterministic exploration
b) Random sampling
c) Breadth-first expansion
d) Backtracking
Correct Answer: b) Random sampling

6. Which AI technique repeatedly samples possible outcomes to estimate the best solution?

a) Hill Climbing
b) Monte Carlo Search
c) Resolution
d) Forward Chaining
Correct Answer: b) Monte Carlo Search

7. Which feature makes Monte Carlo Search suitable for complex decision-making problems?

a) Eliminates uncertainty completely
b) Estimates solutions through repeated simulations
c) Requires no search process
d) Guarantees the shortest path
Correct Answer: b) Estimates solutions through repeated simulations

8. Ontological engineering is mainly concerned with

a) Designing databases
b) Representing domain knowledge formally
c) Developing operating systems
d) Programming robots
Correct Answer: b) Representing domain knowledge formally

9. Which component represents real-world entities in an ontology?

a) Objects
b) Actions
c) Sensors
d) Utilities

Correct Answer: a) Objects

10. What is the primary role of ontologies in knowledge sharing?

a) They provide standardized knowledge representation
b) They reduce memory size
c) They replace inference engines
d) They eliminate uncertainty

Correct Answer: a) They provide standardized knowledge representation

11. Which concept represents occurrences that happen over time?

a) Categories
b) Events
c) Predicates
d) Clauses

Correct Answer: b) Events

12. Which logic is commonly used to represent beliefs, knowledge, and necessity?

a) Boolean Logic
b) Modal Logic
c) Propositional Logic
d) Predicate Logic

Correct Answer: b) Modal Logic

13. Reasoning systems for categories primarily support

a) Object classification
b) Data encryption
c) Image enhancement
d) Process scheduling

Correct Answer: a) Object classification

14. Which reasoning system classifies objects based on their properties?

a) Category Reasoning
b) Resolution
c) Forward Chaining
d) Backtracking

Correct Answer: a) Category Reasoning

15. Default reasoning is mainly used when

a) Complete information is available
b) Information is incomplete or uncertain
c) Arithmetic calculations are performed
d) Database queries are executed

Correct Answer: b) Information is incomplete or uncertain

16. What is the purpose of default reasoning in Artificial Intelligence?

a) Handle incomplete information
b) Eliminate uncertainty completely
c) Replace Bayesian Networks
d) Avoid logical inference

Correct Answer: a) Handle incomplete information

17. Which model represents probabilistic relationships among variables?

a) Bayesian Network
b) Resolution Tree
c) State Space Tree
d) Decision Table
Correct Answer: a) Bayesian Network

18. Which reasoning method allows conclusions based on default assumptions?

a) Resolution
b) Default Reasoning
c) Forward Chaining
d) Backward Chaining
Correct Answer: b) Default Reasoning

19. Which ontology component groups similar objects together?

a) Categories
b) Predicates
c) Rules
d) Events
Correct Answer: a) Categories

20. Which technique is most suitable for decision-making under uncertainty?

a) Bayesian Network
b) Resolution
c) Hill Climbing
d) Breadth First Search
Correct Answer: a) Bayesian Network